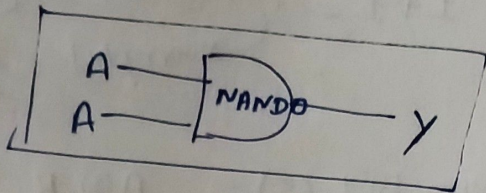
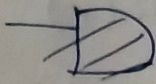
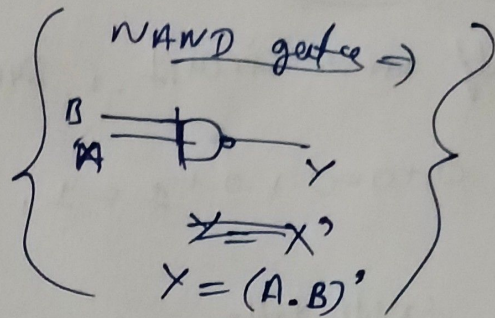


Digital Assignment.

⑨ A NOT gate $\Rightarrow$
using NAND gate

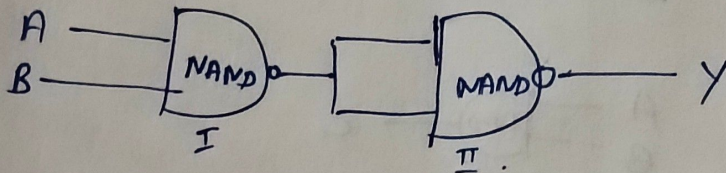


$$Y = (A.A)' = A'$$



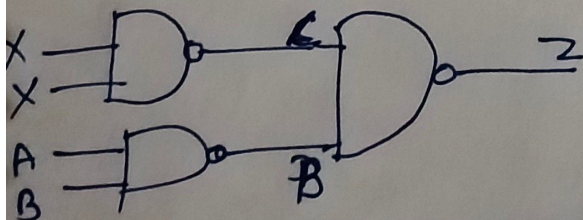
A	Y
0	1
1	0

⑩ AND gate using NAND gate $\Rightarrow$



A	B	Y
1	0	0
0	1	0
0	0	0
1	1	1

⑪ OR gate using NAND gate $\Rightarrow$



$$Z = (A'.B')' = (A')' + (B')' = A + B$$

A	B	X	C	D	Z
1	0	0			
0	1				
0	0				
1	1				

X	A	B	Z
0	0	1	0
1	0	1	1
0	1	0	1
1	1	0	1

Mission - 2

Binary addition $\Rightarrow$

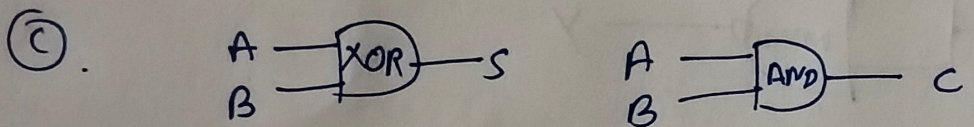
Half adder circuit, Sum bit (S), carry bit (C).

$\Rightarrow 0+0=0, 0+1=1, 1+0=1, 1+1=$ $\begin{cases} \text{Sum bit} = 0 \\ \text{Carry bit} = 1 \end{cases}$

(a) truth table.

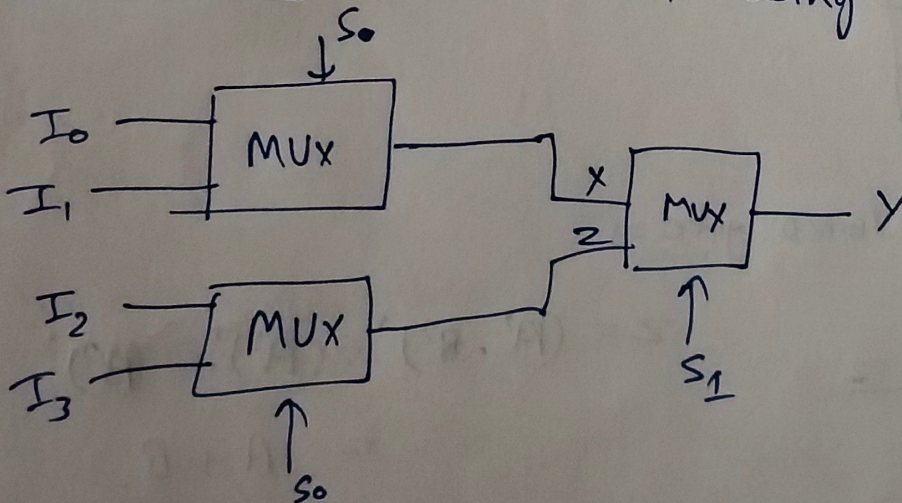
A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	1
1	1	0	1

(b) Sum bit (S) = $A \oplus B$
Carry bit (C) = $A \cdot B$.



Mission - 3

Used Mux using 2 to 1 mux.



truth table

S_1	S_0	Y
0	0	I_0
0	1	I_2
1	0	I_1
1	1	I_3